Amna Zia

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Education

- **Georgia Institute of Technology** — Atlanta, GA
  Bachelor of Science in Mechanical Engineering — *Expected May 2028*
  - Dean's List
  - Coursework: Statics, MATLAB, Linear Algebra, Engineering Graphics, Engineering Materials, Mechanical Design

Experience

- **Data Analytics Externship** — Remote
  NASCAR, Extern — *June 2026 – July 2026*
  - Analyzed NASCAR race and sponsorship data using Python, APIs, and web scraping to identify sponsor visibility trends.
- **AI/ML Research Assistant** — Atlanta, GA
  Flow Physics and Computational Science Lab, Georgia Tech — *Dec 2025 – Present*
  - Generated high-fidelity multiphase turbulence simulations on Aurora for MFlowDB, an open-source dataset built with Argonne National Lab & NVIDIA that replaces costly physical experiments.
  - Set up and validated simulations using fluid mechanics principles to ensure physically accurate outputs.
  - Benchmarked deep learning models (autoencoders, neural operators) for predicting flow evolution across turbulence regimes, and used SHAP to identify the key flow features driving predictions.
- **Research Lab Intern** — Atlanta, GA
  Center for Research and Therapeutic Development, Clark Atlanta University — *May 2023 – July 2023*
  - Assisted senior researchers with experimental setup, data collection, and sample preparation for cancer research in molecular cell biology.
- **FRC Robotics**
  Lead Team Member — *Sep 2021 – May 2024*
  - Led design and development of competition robots, integrating mechanical subsystems with programming and controls.
  - Mentored a team of 10+ members, building programming and design skills while promoting effective collaboration.
  - Led the team to notable achievements in regional and state robotics competitions.

Projects

- **Autonomous Arena Mechatronic System – Mechanical Design Project** — Atlanta, GA
  Mechanical Design Course, Georgia Institute of Technology — *Aug 2026 – Present*
  - Designed and built an autonomous robot with a motor-driven drivetrain, pneumatic flapper, and solenoid-actuated release door.
  - Programmed a Teensy in C to sequence and execute all three arena tasks.
- **Row-Weeding Robot – Drivetrain CAD Design** — Atlanta, GA
  Mechanical Design Project, Georgia Institute of Technology — *Jan 2025 – May 2025*
  - Modeled a multi-part wheel subsystem in SolidWorks for an autonomous agricultural robot, including drivetrain components, assemblies, and motion constraints.
  - Improved fit, stability, and manufacturability through iterative CAD, tolerance analysis, and chassis integration.
- **Crescent Lantern – Product Design Project** — Atlanta, GA
  3D Mechanical Design Project, Georgia Institute of Technology — *Jan 2025 – May 2025*
  - Designed a multifunctional lantern in SolidWorks that combines decorative aesthetics with a functional tea light candle holder using iterative 3D-printed prototypes.
  - Improved product functionality and manufacturability through iterative CAD modeling and prototyping.
- **Girls' Education**
  Programming Website Project — *Feb 2024 – March 2024*
  - Built a responsive website (HTML, CSS, JavaScript) raising awareness of girls' education inequality, with a donation feature supporting the Malala Fund.

Skills

- **CAD & Design**: SolidWorks (parts, assemblies, drawings), GD&T, DFM/DFA, Fits & Clearances
- **Prototyping**: 3D printing, Iterative prototyping, Hands-on assembly, Laser cutting
- **Programming**: MATLAB, Python, C, HTML/CSS
- **Software**: Microsoft Excel, Word, PowerPoint, Outlook

Activities

- **Society of Women Engineers (SWE), Georgia Tech**
  Member — *Aug 2024 – Present*
- **Junior STEM, Georgia Tech**
  Volunteering Member — *Jan 2025 – Present*
- **HyTech Racing, Georgia Tech**
  Mechanical Engineering Team — *Aug 2026 – Present*